Investigating the Efficacy of Simulation Coding Exercises for Teaching Probability Theory

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Background

- **Current Probability Courses:** Traditional probability courses tend to emphasize symbolic derivations and hand-written problem sets, which build algebraic skills but may not help students develop a clear mental picture of random processes
- **New Approaches:** Statistics and data science courses have increasingly adopted simulation-based, coding-centered approaches, where learners write short programs, generate many samples from a model, and use plots and summaries to "see" theoretical ideas in action [1]

Two Hypotheses

1. Whether access to coding exercises, in any form, leads to additional learning gains beyond traditional materials alone.
2. Whether combining coding with handwritten work yields benefits beyond using either modality by itself.

Methods

Developed multiple exercise sets on a variety of topics then selected one for testing

Assigned 50 participants to one of four groups: assessment only, coding + assessment, handwritten questions + assessment, or both modalities.

Raw intake and assessment data were cleaned and standardized, normalized and confidence-adjusted scores were computed, and finalized datasets were generated for analysis.

During EDA, raw and confidence-adjusted scores were analyzed by treatment group and demographic information.

Fit 2 models to test our hypotheses.

Two Main Models

1. $\beta 0 + \beta 1(\text{coding}) + \beta 2(\text{handwritten}) + \beta 3(\text{interaction})$
2. $\beta 0 + \beta 1(\text{coding} + \text{handwritten}) + \beta 3(\text{interaction})$

References

[1] Rossman & Chance - Using Simulation-Based Inference for Learning Introductory Statistics https://doi.org/10.1002/wics.1302

Results

Unadjusted assessment scores

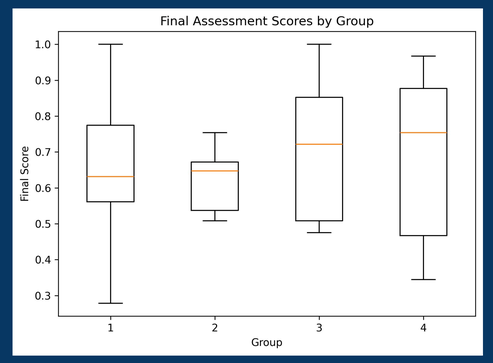

Figure 1: Final assessment score distributions by group

There was no statistically significant effect of coding, handwritten practice, or their interaction on final assessment scores

With a p-value of 0.886, we fail to reject the null hypothesis at the 0.05 level. There is insufficient evidence that combining coding and handwritten questions improves assessment scores.

	Coef.	Std.Err.	t	P>|t|	[0.025	0.975]
Intercept	0.629326	0.048367	13.011567	5.027419e-17	0.531969	0.726683
coding	0.001821	0.076474	0.023818	9.811005e-01	-0.152114	0.155756
handwritten	0.070128	0.083774	0.837108	4.068598e-01	-0.098500	0.238755
coding:handwritten	-0.017221	0.119812	-0.143736	8.863365e-01	-0.258391	0.223948

Figure 2: model fit to predict final assessment scores

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	47.0	1.960964	0.0	NaN	NaN	NaN
1	46.0	1.936968	1.0	0.023995	0.569846	0.454168

Figure 3: Results from ANOVA test comparing models 1 and 2

With a p value of 0.453, we failed to reject the null hypothesis at the 0.05 level. There is insufficient evidence that coding exercises would increase assessment scores.

Results

Adjusted assessment scores

There was no statistically significant effect of coding, handwritten practice, or their interaction on final assessment scores adjusted for confidence

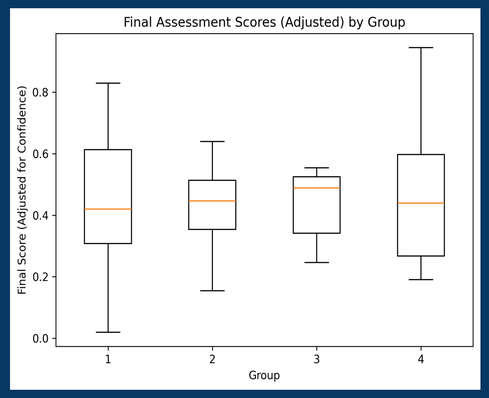

Figure 4: Final assessment adjusted score distributions by group

	Coef.	Std.Err.	t	P>|t|	[0.025	0.975]
Intercept	0.434426	0.049614	8.756188	2.330031e-11	0.334559	0.534293
coding	-0.012295	0.078446	-0.156733	8.761412e-01	-0.170199	0.145609
handwritten	0.027140	0.085933	0.315829	7.535607e-01	-0.145835	0.200115
coding:handwritten	0.019879	0.122901	0.161749	8.722122e-01	-0.227508	0.267267

Figure 5: model fit to predict adjusted final assessment scores

With a p-value of 0.872, we fail to reject the null hypothesis at the 0.05 level. There is insufficient evidence that combining coding and handwritten questions improves assessment scores.

With a p value of 0.673, we failed to reject the null hypothesis at the 0.05 level. There is insufficient evidence that coding exercises would increase assessment scores.

	df_resid	ssr	df_diff	ss_diff	F	Pr(>F)
0	47.0	2.046130	0.0	NaN	NaN	NaN
1	46.0	2.038132	1.0	0.007998	0.18051	0.672916

Figure 6: Results from ANOVA test comparing models 1 and 2

Adding in Factors from Intake Form

We added the following features from our intake form into the model to see if they helped with final assessment score prediction.

- Affiliation with UCSD Data Science
- Year at UCSD
- Familiarity with statistics ranked 1-5

Only a small number of intake form features show significant relationships with final assessment scores, suggesting limited improvement in prediction.

	Coef.	Std.Err.	t	P>|t|	[0.025	0.975]
Intercept	0.424988	0.078878	5.387943	0.000005	0.265017	0.584960
DSC_major	-0.171539	0.078396	-2.188111	0.035234	-0.330534	-0.012545
Year_4	0.099106	0.067718	1.463514	0.152004	-0.038232	0.236443
Stats_3	0.030567	0.077864	0.392566	0.696955	-0.127349	0.188483
Year_1	0.068492	0.058416	1.172486	0.248702	-0.049981	0.186966
Year_transfer_1	-0.101551	0.072552	-1.399701	0.170162	-0.248692	0.045591
handwritten	0.056697	0.078346	0.723677	0.473941	-0.102195	0.215589
Year_transfer_2	0.234096	0.100089	2.338882	0.025010	0.031106	0.437085
Year_2	0.004956	0.077406	0.064024	0.949305	-0.152032	0.161943
DSC_minor	-0.219788	0.082015	-2.679862	0.011037	-0.386121	-0.053455
Stats_1	0.123414	0.113365	1.088644	0.283545	-0.106500	0.353328
Stats_2	0.162814	0.087471	1.861350	0.070876	-0.014585	0.340214
Year_3	0.119889	0.067525	1.775481	0.084273	-0.017058	0.256837
coding_handwritten	0.040992	0.111120	0.368898	0.714364	-0.184370	0.266353
coding	0.010626	0.078625	0.135145	0.893250	-0.148832	0.170084

Figure 7: model fit to predict final assessment scores using features from intake form

Limitations & Future Work

- Our low sample size(50) was the biggest limitation in generating any significant findings.
- We are going to continue gathering data in an effort to publish our findings
- The barrier to entry of our exercises was knowledge of python, and more research could be done on helping students with no coding knowledge run simulations.

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